Multiscale Feature Knowledge Distillation and Implicit Object Discovery for Few-Shot Object Detection in Remote Sensing Images

Jie Chen, *Member, IEEE*, Ya Guo, Dengda Qin, Jingru Zhu, Zhenbo Gou, and Geng Sun

***Abstract*— Dynamic or sudden changes in various scenes may give rise to new objects. These new objects with limited annotated samples are susceptible to overfitting in deep learning. While few-shot object detection (FSOD) is effective with limited samples, current FSOD methods for remote sensing images still face specific challenges. The "pretraining-transfer" paradigm tends to forget the feature representations of base classes, impacting the learning process for novel classes during few-shot training. Furthermore, the presence of implicit objects in sparsely labeled instances of remote sensing images introduces erroneous supervisory information. To address these challenges, we propose an FSOD method that incorporates multiscale feature knowledge distillation and implicit object discovery, named MFKDIOD, which preserves the performance of base classes and mitigates the impact of implicit objects. Specifically, we first design a multiscale feature knowledge distillation (MFKD) module, which transfers the knowledge of base classes from a teacher network to a student network, enabling the student network to better retain the base class feature representations. Second, we design an implicit object discovery (IOD) module that utilizes both the teacher and student networks to discover implicit objects within the few-shot training data and generate pseudolabels. The code will be available at https://github.com/RS-CSU/MFKDIOD.**

***Index Terms*— Few-shot learning, knowledge distillation, object detection, pseudolabel.**

I. Introduction

OBJECT detection aims to classify and locate the interested objects in remote sensing images [1]. It is a fundamental task in remote sensing image analysis and understanding, with broad applications. With the development of sensor technology, the spatial resolution of remote sensing images has been increasing, and people are able to observe more fine-grained objects [2], such as airplanes, cars, ships, or smaller wind turbines. In addition, the Earth's surface undergoes rapid and diverse changes, leading to the emergence of novel objects in various geographic scenes. For instance, urban disaster scenarios require identifying damaged buildings. The combination of high spatial resolution and dynamic scene changes may lead to the emergence of new objects, which causes the problem of having a limited number of samples available for object detection.

In recent years, deep neural networks (DNNs) have exhibited strong feature extraction capabilities, leading to a series of high-performance object detection algorithms [3], [4], [5], [6], [7]. Recently, the emergence of remote sensing large models such as RingMO [8], RSPrompter [9], and RemoteCLIP [10] has further enhanced the feature extraction and generalization capabilities of object detection models. However, most object detection algorithms heavily rely on a large amount of annotated data to obtain robust feature representations. In the remote sensing few-shot object detection (FSOD) task, new objects often have limited samples available, which can lead to overfitting in DNN-based object detection methods. This is because the scarcity of training samples makes it difficult for the models to learn the feature representations of novel classes [11]. In addition, due to the diversity of target orientations coming from sensor angles and the diversity of remote sensing object scales [12], the general FSOD methods that are designed for natural images may not be suitable for remote sensing images.

Currently, a series of outstanding FSOD methods are proposed to address the feature extraction problem under limited samples. Existing FSOD methods can be divided into three categories: 1) fine-tuning-based methods [13], [14], [15]; 2) metric learning-based methods [16], [17]; and 3) model improvement-based methods [18], [19], [20]. Fine-tuning-based methods are primarily based on the idea of freezing most of the parameters of the backbone network and fine-tuning only a few parameters relevant to the detection task on few-shot annotations. For example, TFA [13] is a typical fine-tuning-based FSOD method that performs few-shot training by freezing the main model parameters. However, the method tends to forget discriminative features of the base classes when fine-tuning on few-shot datasets and lacks generalization on the novel classes. Metric learning-based methods aim to construct a metric space to improve the accuracy of classification. For example, RepMet [16] classifies objects by computing the distances between embedding space vectors and class prototypes. However, when the interclass variance is large, the category centers computed in the metric space using the few-shot annotation do not represent the class prototype. In recent times, model improvement-based methods

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The authors are with the School of Geosciences and Info-Physics, Central South University, Changsha 410083, China (e-mail: sungeng@csu.edu.cn).

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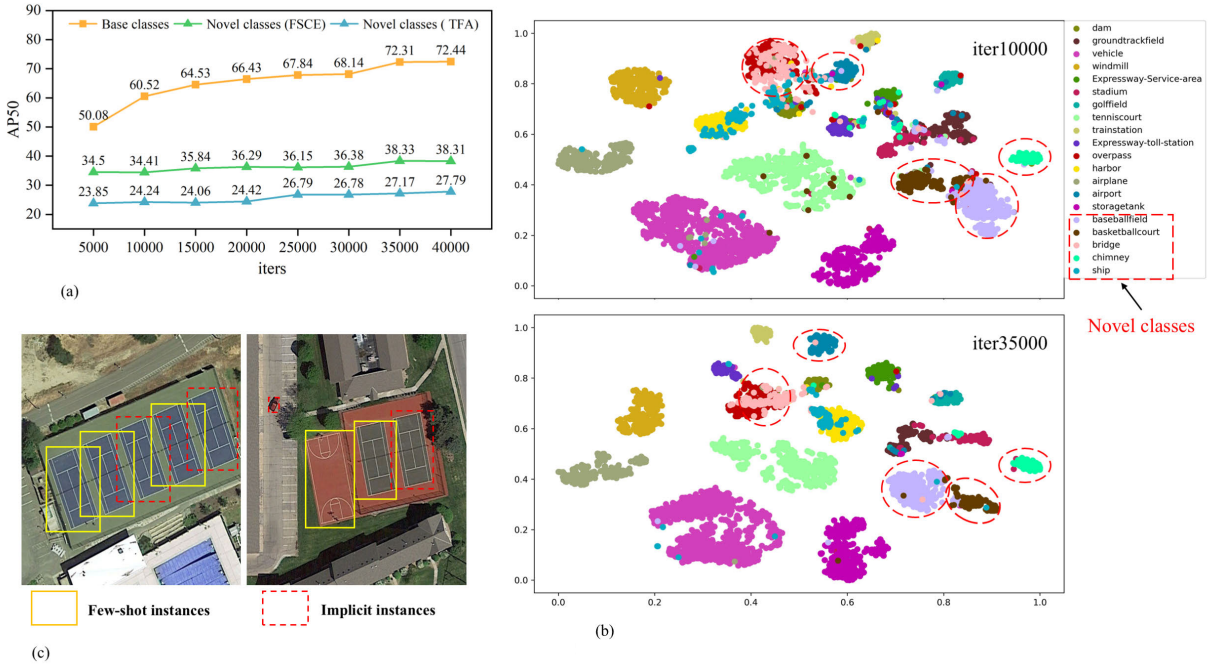


Fig. 1. Effect of different pretraining models on the novel classes and presence of implicit objects. (a) Accuracy of novel and base classes when loading different iterations' pretraining weight. (b) Visualization of novel and base class features of FSCE when loading different iterations' pretraining weight, novel classes in red ellipse. (c) Presence of implicit objects under the few-shot task setting.

have gained prominence. These methods focus on optimizing the model's detection capabilities by incorporating auxiliary branches, thereby allowing for more effective utilization of the limited available samples. For example, FSCE [20] adds a contrast learning branch to the few-sample detector to alleviate the misclassification problem.

Although model improvement-based methods have achieved significant success, there are still two issues to consider: 1) the problem of forgetting the base class feature representations and 2) the problem of wrong supervision due to implicit objects in few-shot images. To understand Problem 1, we conducted a simple exploratory experiment using these two classical FSOD methods. TFA and FSCE first pretrain on the base classes dataset with different iterations and produce the accuracy on the base classes; then, the few-shot detector is loaded with the pretrained model weights generated with different iterations for enough few-shot training (3000 iterations) and produce the accuracy on the novel classes [as shown in Fig. 1(a)]; and finally, for the FSCE loaded with the pretrained model weights generated with 35 000 iterations and 10 000 iterations of pretraining, visualize the feature distribution of all classes after few-shot training (using t-SNE dimensionality reduction), as shown in Fig. 1(b). We have the following findings: 1) the accuracy of the novel classes is highly correlated with whether or not the base classes are adequately trained, and the maximum accuracy difference is about 4% under the FSCE method and 2) the more adequately trained base classes model is, the more distinguishable are the features of the novel classes and the base classes after few-shot training, as shown in Fig. 1(b). We believe that the stronger the feature representation ability of the base classes model is, the more favorable it is for the learning of the novel classes. In other words, the performance of the novel classes is positively correlated with the performance of the base classes. However, the few-shot training process of the model is prone to the forgetting of the base class feature representation in the "pretraining transfer" paradigm. This phenomenon may not be conducive to few-shot learning. For Problem 2, due to the high density and multiclass coexistence of objects in remote sensing images, there will be some implicit objects in the image under the few-shot annotation, as shown in Fig. 1(c). If they are not additionally processed, they are treated as background in the few-shot training, thus providing wrong supervised information. LVC [21] has explored this problem and used pseudolabeling techniques to mitigate it. However, LVC only performs one iteration of pseudolabeling for the candidate object boxes of novel classes using a kNN classifier. The quality of the pseudolabels heavily relies on the feature extraction capability of the ViT [22] model (pretrained on the ImageNet-10 dataset) and the candidate boxes generated by the RPN network. Once there are omissions and errors in the pseudolabels, they will continue to provide wrong supervision throughout the subsequent end-to-end training. In addition, the ViT weights are trained from the natural image dataset ImageNet-10, which is not suitable for feature representation of remote sensing objects. In remote sensing images, due to the high density of objects, the issue of implicit objects becomes even more significant.

To address the mentioned challenges, this article proposes a novel FSOD method called multiscale feature knowledge distillation and implicit object discovery (MFKDIOD). More specifically, we propose the multiscale feature knowledge distillation (MFKD) module to address the problem of forgetting the feature representations of base classes. The MFKD module

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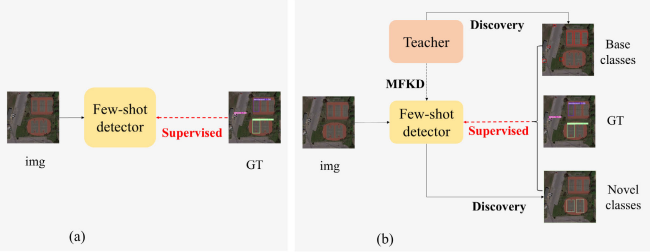


Fig. 2. General FSOD method and the teacher–student structure of our proposed method. (a) General FSOD method. (b) Our teacher–student method.

leverages the multiscale feature knowledge extracted from both the teacher network and the student network. It uses MFKD loss to transfer the base class feature representation to the student network, thus boosting the learning of the novel classes in a few-shot scenario. The MFKD module allows the few-shot detector to learn base class features from the teacher network, thus alleviating the forgetting of base class knowledge. In addition, to contend with the influence of implicit objects, we propose the implicit object discovery (IOD) module. The IOD module utilizes the teacher network to discover implicit objects of the base classes and generate pseudolabels for them. It also employs the student network to discover implicit objects of novel classes in the images. The discovered implicit objects are subsequently integrated with the ground truth to construct the training labels for the student network. The IOD module simultaneously discovers implicit objects of both base and novel classes in the training, leading to the acquisition of more comprehensive supervision information.

The architecture of the proposed method is different from the general few-shot method, as shown in Fig. 2. For general few-shot fine-tuning methods (taking TFA as an example), the network is first pretrained on the base classes dataset, and then, the pretrained weights are loaded for fine-tuning on the few-shot dataset [see Fig. 2(a)]. In this pretraining-transfer learning paradigm, the network gradually overfits the novel classes and forgets the features of the base classes. The proposed method employs a teacher–student architecture, using knowledge transfer between the teacher and student networks (MFKD module) to alleviate base class feature forgetting. In addition, general few-shot fine-tuning methods directly fine-tune on datasets containing K (=3, 5, 10, 20) instances of novel and base classes samples (GT), where unannotated instances of novel and base classes may cause misguidance in updating network parameters. In contrast, during our few-shot training phase, the labels include automatically annotated implicit base and novel class instance objects [as shown in Fig. 2(b)].

1) We proposed a novel remote sensing image FSOD method, MFKDIOD, which aims to solve the problems of base class feature representation forgetting and implied objects. Experiments on the DIOR dataset and the NWPU VHR-10 dataset have proved the advancement of the proposed method.
2) We proposed the MFKD module, which slows down the forgetting of base class features through multiscale knowledge distillation, and at the same time, the stronger base class feature representation facilitates the learning of novel class features.
3) To reduce the interference of implicit objects, we proposed the IOD module, which discovers the implicit objects of the base classes and novel classes with limited samples and provides more comprehensive supervised information.

II. Related Works

A. Object Detection

In recent years, object detection methods based on deep features have made significant progress, thanks to the powerful feature extraction capabilities of convolutional neural networks. Based on whether the network includes region proposal structures, current object detection methods can be divided into two-stage detection and one-stage detection. Two-stage detection consists of two stages: the first stage generates region proposals, and the second stage performs precise localization and classification using these proposals. For instance, Faster R-CNN [23] introduced the region proposal network (RPN) to replace selective search in Fast R-CNN [24]. This method first generates region proposals using the RPN and then refines them through regression and classification. Methods based on region proposals have achieved excellent performance. Many researchers have improved on this foundation, such as Mask R-CNN proposed by He et al. [25], which integrates segmentation tasks into object detection to enhance detection performance. One-stage detection methods directly predict the bounding box coordinates and classification labels using convolutional neural networks. The YOLO series [3], [4] represents typical one-stage object detection methods. To eliminate the need for predefined anchor boxes in detection, Law and Deng first proposed CornerNet [26], an anchor-free object detection method that further improves speed and accuracy. CenterNet [27] also adopts an anchor-free approach, using key points as the basis for detection. FCOS [28] performs anchor-free detection by representing bounding boxes with anchor points, lengths, and widths. Recently, transformer-based methods [29] have gained attention due to their stronger image representation capabilities. For example, DETR [29] performs direct end-to-end detection based on transformers [30] without postprocessing.

The above methods have shown some effectiveness in remote sensing images. However, remote sensing images have complex backgrounds and different scales for the objects [31]. Hence, in response to the challenges presented by remote sensing images, many remote sensing object detection methods have been proposed. For example, Chen et al. [5] incorporated multiscale spatial and channel attention into the object detection network to improve the model's focus on the objects of interest. Cui et al. [32] used a grouped attention mechanism in ship detection to increase the model's attention to ship objects. To deal with the different scales of objects in remote sensing images, Deng et al. [6] constructed a multiscale object proposal network (MS-OPN) and an accurate object detection network (AODN). To address the challenges of scale variation

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and intraclass similarity, CSFF [7] uses FPN and squeeze and excitation (SE) modules for object detection. MFENet used a global feature perception (GFP) module to extract global multiscale features of salient object regions [33].

B. Few-Shot Object Detection

In recent years, FSOD has gained attention for its ability to detect novel category objects with limited samples. Currently, FSOD methods can be categorized into three types: 1) fine-tuning-based methods [13], [14], [15]; 2) metric learning-based methods [16], [17]; and 3) model improvement-based methods [18], [19], [20]. Fine-tuning-based methods are primarily based on the idea of freezing most of the parameters of the backbone network and fine-tuning only a few parameters relevant to the detection task on few-shot samples. For example, TFA [13] performs few-shot training by freezing the main model parameters. FADI [15] associates the novel classes with base classes that have similar semantics and then decomposes the associated model into base and novel class branches for fine-tuning. Metric learning-based methods aim to construct a metric space to improve the accuracy of classification. For example, RepMet [16] classifies objects by computing the distances between embedding space vectors and class prototypes. To enhance the separability between positive and negative feature representations, Yang et al. [17] improve few-shot detection performance by introducing a negative sample metric space in RepMet. Model improvement-based methods typically use separate conditional branches to assist in enhancing the detection capabilities of the model. For example, Meta RCNN [18] combines the concept of metalearning and reshapes the R-CNN prediction head using channelwise attention. Fsdet [19] aggregates the output features of the class encoder and query features with reweighting to improve the confidence of few-shot detection. MSPR [34] refines the features of candidate boxes using a combination of feature pyramid networks and image pyramids, with the instance objects cropped as input. FSCE [20] uses supervised contrastive learning as an auxiliary task to mitigate the problem of inaccurate classification.

In remote sensing images, sudden changes or rapid development of cities generate new objects of interest that have a small number of labeled samples. Therefore, many methods have started to focus on the research of FSOD specifically for remote sensing images. Li et al. [35] improved FSOD in remote sensing images by using an elemental feature extractor to generate three different sizes of elemental features and applying feature reweighting on the YOLO v3 network architecture. Cheng et al. [36] addressed the problems of directionally sparse and spatially scattered objects in few-shot remote sensing object detection by proposing P-CNN to obtain better object prototypes and generate high-quality candidate boxes from complex backgrounds, thereby improving detection accuracy. DMMLNet [37] employed the deep feature pyramid comparison network (DFPCN) as a multiscale metric learning architecture and introduced an affinity fusion mechanism to address the task of few-shot instance segmentation. MSOCL [11] optimized the detection module by using contrastive learning to assist in cropping instance objects and improving the model's feature representation. MM-RCNN [38] used memory module and external attention mechanisms to improve the detection module by taking into account the interclass relationships between object categories in remote sensing images.

Although some studies have considered the phenomenon of overfitting [13], the effect of the forgetting of the base class feature representations on the novel classes should also be considered. Furthermore, although current FSOD methods can enhance the utilization of limited training data, there is still a need for deeper exploration of the problem of erroneous supervision caused by implicit objects in the few-shot training data. Therefore, further research is needed to advance the field of FSOD in remote sensing by addressing these challenges.

C. Knowledge Distillation

Knowledge distillation is a concept introduced by Hinton et al. [39] and is an effective method for model compression. It uses the soft targets from a teacher network as part of the supervision to guide the training of a student network and transfer the knowledge from the teacher network [39]. Knowledge distillation has found wide applications in model compression and quantization. For example, in [40], knowledge distillation is used to compress object detection models and improve detection accuracy. Wang et al. [41] proposed feature imitation to effectively extract knowledge from the teacher network for network quantization. As research progresses, knowledge distillation has been applied to address other issues in models. For example, Feng et al. [42] used knowledge distillation to solve catastrophic forgetting in incremental object detection. Guo et al. [43] applied the idea of knowledge distillation to mitigate the impact of pseudolabel noise in semisupervised object detection. Chen et al. [44] investigated the problem of incremental object detection in remote sensing images using knowledge distillation. Similarly, knowledge distillation has also been employed in methods for FSOD. For example, DID [45] treated novel class detection as a new task to solve the forgetting problem caused by training on the new task. MDFC [46] used a memory bank and multifaceted distillation to support few-shot learning.

Analyzing these studies, it can be observed that knowledge distillation enables the student network to "mimic" the teacher network, allowing the student network to preserve certain characteristics of the teacher network. Drawing inspiration from knowledge distillation, it is possible to use the student network to learn the feature representations of the teacher network and to address the problem of feature degradation in FSOD. Differing from the methods referenced earlier, our distillation module aims to distill the base class feature representation of the teacher network, thus enhancing the learning of novel classes more effectively.

III. Methodology

In this section, we will present the problem setting for FSOD and provide an overview of the proposed method. Our approach incorporates MFKD and the discovery of implicit objects. Subsequently, we will introduce the essential

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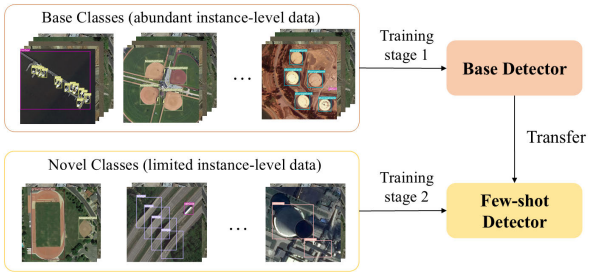


Fig. 3. Pretraining-transfer paradigm for the FSOD framework.

modules of the proposed method, specifically, teacher–student architecture, MFKD, and IOD.

A. Problem Formulation

Our task setting is similar to prior studies [11], [36]. It involves two datasets, namely, the base classes dataset (D_{base}) and the novel classes dataset (D_{novel}), as shown in Fig. 3. D_{base} contains abundant instance-level annotated data for pretraining models. The classes within D_{base} belong to the base classes, denoted as C_{base}. Conversely, D_{novel} contains instance-level annotated data for few-shot training. The classes within D_{novel} are denoted as C_{novel}. The few-shot data are commonly expressed as K-shot, indicating the number of training instances available per class.

The training of FSOD tasks generally follows the pretraining-transfer paradigm [47], as shown in Fig. 3. Specifically, the base detector is pretrained using D_{base} in training stage 1. Then, the few-shot detector is trained using D_{novel} by transferring the feature knowledge of the base detector in training stage 2. Usually, the final FSOD model is designed to better detect the novel classes while maintaining the performance of the base classes to the greatest extent possible.

B. Overall Framework

The overall framework of the proposed method for FSOD, incorporating MFKD and IOD module, is shown in Fig. 4. This approach is designed to address the problem of forgetting base class feature representations and the negative impact of implicit objects in FSOD. Specifically, the proposed method is based on the teacher–student architecture, where the MFKD module ensures that the base class feature representation capability of the teacher network is distilled into the student network. This aims to slow down the decay of feature representation capability during few-shot training and preserve the feature representation capability of the few-shot detection method. Furthermore, the IOD module is employed to identify implicit objects within the images and address the problem of erroneous supervision stemming from them. This module uses both the teacher and student networks to discover implicit objects within the few-shot samples. The detected objects are then subjected to postprocessing to generate pseudolabels for the implicit objects. Finally, the real labels are fused with the pseudolabels to generate the training labels used in the few-shot training process.

For the training of MFKDIOD, the first stage uses the base class dataset to update the parameters of the teacher network. In the second stage of training, the parameters of the teacher network are frozen, and the parameters of the student network are updated using the K-shots instances in the image, as shown in Fig. 4. For the teacher and student networks, a more detailed description is given in Section III-C.

C. Teacher–Student Architecture

The teacher–student architecture is the foundation for performing MFKDIOD. In our method, the teacher network provides the knowledge of multiscale features from the base classes and the pseudolabels of implicit objects in the base classes. The student network receives the knowledge from the base classes and discovers the pseudolabels of implicit objects in the novel classes while completing the few-shot training.

1) Teacher: To effectively transfer the multiscale feature knowledge from the teacher network to the student network, we employ base classes dataset pretrained model as the teacher network. In our approach, the teacher network uses the Faster RCNN network [23]. The teacher network has been trained with a large number of base class objects and has good generalization performance in detecting base class objects. During few-shot training, the parameters of the teacher network are frozen to provide multiscale feature knowledge and pseudolabels of base class implicit objects for the student network, as shown in the dashed box in Fig. 4.

2) Student: During the few-shot training phase, the student network uses MSOCL [11] that we proposed in our previous work. MSOCL is a Siamese network structure modified from Faster RCNN, which enhances the feature representation for FSOD through multiscale object contrastive learning, as shown in the solid box in the lower right corner of Fig. 4. A more specific introduction to student networks can be found in our previous work. MSOCL contains the various modules of Faster RCNN and object contrastive learning. Therefore, the loss function of MSOCL can be expressed as follows:

$$\mathbb{L} = \mathcal{L}_{\text{RPN}} + \mathcal{L}_{\text{cls}} + \mathcal{L}_{\text{reg}} + \lambda \mathcal{L}_{(y^o, y^p)} \tag{1}$$

where $\mathcal{L}_{\text{RPN}}$ represents the loss of the RPN, $\mathcal{L}_{\text{cls}}$ represents the cross-entropy loss for the classification of proposals, $\mathcal{L}_{\text{reg}}$ represents the smooth $L1$ loss for computing the bounding box loss, $\mathcal{L}_{(y^o, y^p)}$ is the contrastive multiscale proposal (CMSP) loss [11], and λ is set to 0.2 in our experiments to balance the CMSP loss.

During the few-shot training stage, the student network receives the multiscale feature knowledge from the teacher network and learns to detect novel class objects. Specifically, images containing K instances of the base class and the novel class are fed into the teacher network and the student network, respectively, and the parameters of the teacher network are frozen, while the parameters of the student network are updated. First, the teacher network extracts multiscale features and passes them to the student network through the MFKD module, which is described in Section III-D. Second, the teacher network discovers the implicit objects of the base classes and processes them to generate pseudolabels y_t.

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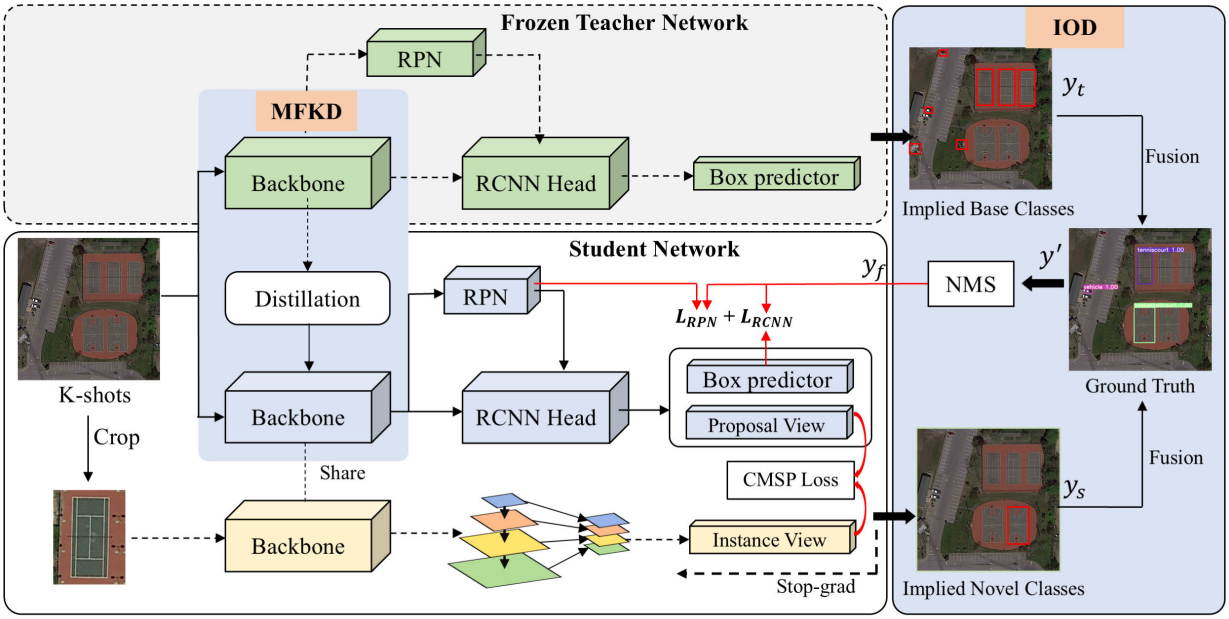


Fig. 4. Overall architecture of MFKDIOD, where MFKD is the multiscale feature knowledge distillation module and IOD is the implicit object discovery module. Red arrows indicate the loss of the student network, and dashed arrows indicate no gradient back propagation in the few-shot training.

The student network starts to discover the implicit objects of the novel classes after 500 warm-up iterations and generates pseudolabels y_s. Finally, the pseudolabels for the base classes y_t and the pseudolabels for the novel classes y_s with the ground-truth labels y' are fused to obtain the fused labels y_f, which provide supervision for the student network.

D. Multiscale Feature Knowledge Distillation

The knowledge used for distillation can be categorized into response-based knowledge [40], relation-based knowledge [48], and feature-based knowledge [49]. Response-based knowledge relies on the output responses of the teacher network. Relation-based knowledge is built upon the relationships between features or data instances. Feature-based knowledge uses the intermediate layer outputs of both the teacher and student networks as the source of knowledge [50]. Considering that the teacher network can only predict the base classes, while the student network needs to detect both base and novel classes, there is a discrepancy in the number of output responses between the two. Furthermore, the feature representation capabilities of the teacher network for the base classes are mainly embedded in the feature layers, and the decay of feature representation capabilities is primarily related to the parameter updates in the feature layers. Therefore, feature-based knowledge is more suitable for addressing the feature representation decay problem in FSOD.

The structure of the MFKD is shown in Fig. 5. It mainly includes the process of extracting multiscale feature knowledge from the teacher network and obtaining multiscale features from the student network. During the few-shot training, the teacher network uses the feature pyramid network (FPN) to extract multiscale features, which serve as soft labels for the student network. Simultaneously, the student network extracts multiscale features from images,

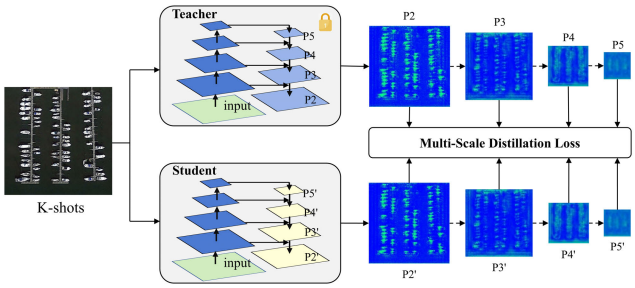


Fig. 5. Structure of the MFKD module. The yellow lock indicates that the parameters of the teacher network are frozen.

providing soft predictions. The MFKD aims to supervise the soft predictions of the student network using the soft labels from the teacher network. The extracted feature knowledge from the teacher network is denoted as y_i^T, where i (2, 3, 4, 5) corresponds to the P2, P3, P4, and P5 features extracted from the FPN. The features are softened by the distillation temperature parameter T and converted into soft labels using the softmax function. The feature extraction process of the teacher network can be represented as follows:

$$y_i^T = \frac{\exp\left(\frac{y_{i,m}}{T}\right)}{\sum_{m=1}^{W \cdot H} \exp\left(\frac{y_{i,m}}{T}\right)} \tag{2}$$

where m represents the number of pixels in the feature map, H and W are the height and width of the feature map, and i represents different scales of feature layers. Similarly, the soft predictions generated by the student network can be represented as

$$y_i^S = \frac{\exp\left(\frac{y'_{i,m}}{T}\right)}{\sum_{m=1}^{W \cdot H} \exp\left(\frac{y'_{i,m}}{T}\right)} \tag{3}$$

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where y_i^S represents the soft predictions of the student network, $y'_{i,m}$ represents the mth pixel in the ith layer of features, and the student network's features i are denoted as P2′, P3′, P4′, and P5′.

To ensure that the student network maintains the feature representation capacity of the base classes, we design an MFKD loss function. Specifically, for each scale of multiscale knowledge, the student network generates soft predictions corresponding to the scale of soft labels. The alignment between soft predictions and soft labels at each scale is quantitatively assessed using KL divergence, which measures the probabilistic divergence between them. In addition, taking inspiration from the findings in the work [51], the MFKD loss is multiplied by y_i^T before the KL divergence. The final expression of this loss is provided in the following:

$$\varphi\left(y^T, y^S\right) = \frac{T^2}{N \cdot C} \sum_{i=1}^{N} \sum_{c=1}^{C} y_i^T \cdot \log\left(\frac{y_i^T}{y_i^S}\right) \tag{4}$$

where $\varphi(y^T, y^S)$ denotes the MFKD loss, N denotes the number of multiscale features, i represents different scales of features, and C denotes the number of feature channels. In addition, the loss calculation includes multiplying the extracted soft objects from the teacher network before the KL divergence. When y_i^T is relatively large, the optimization objective of the loss is to increase y_i^S, while multiplying by y_i^T requires minimizing the KL divergence. On the other hand, when y_i^T is small, the multiplication factor y_i^T can be neglected, and the KL divergence does not need to be strictly minimized. This asymmetric form proves to be more advantageous for knowledge distillation [51].

In the end, the MFKD loss transfers the multiscale feature knowledge in the teacher network to the student network, enabling the student network to have more consistent feature representations with the teacher network on the base classes. The final loss function of the student network consists of two parts: the MFKD loss and the few-shot task loss. The overall loss $\mathbb{L}$ can be formulated as follows:

$$\mathbb{L} = \mathcal{L}_{\text{RPN}} + \mathcal{L}_{\text{cls}} + \mathcal{L}_{\text{reg}} + \lambda \mathcal{L}_{(y^o, y^p)} + \alpha \varphi(y^T, y^S) \tag{5}$$

where $\mathcal{L}_{\text{RPN}}$, $\mathcal{L}_{\text{cls}}$, $\mathcal{L}_{\text{reg}}$, and $\mathcal{L}_{(y^o, y^p)}$ represent the original losses of the student network and α is the weight for the MFKD loss.

E. Implicit Object Discovery

Due to the presence of implicit objects, few-shot models often struggle with distinguishing between foreground and background. Pseudolabeling refers to the model's prediction of unlabeled data to generate labels that can be used for supervision and is often used in semisupervised tasks [52], [53]. For example, Zhou et al. [54] utilized pseudolabels in semisupervised object detection, and Inoue et al. [55] employed a well-trained detector to generate pseudolabels for unlabeled images in cross-domain weakly supervised object detection. These studies demonstrate that pseudolabels can capture implicit information in images, and incorporating object information from extra unlabeled data can effectively guide the model training. With this understanding, implicit objects within few-shot training data can be discovered by the model and used to generate pseudolabels. In the FSOD task, the training dataset includes both base and novel classes, and not all object classes can be discovered by a single model. Therefore, the process of discovering implicit objects is primarily divided into base class object discovery and novel class object discovery.

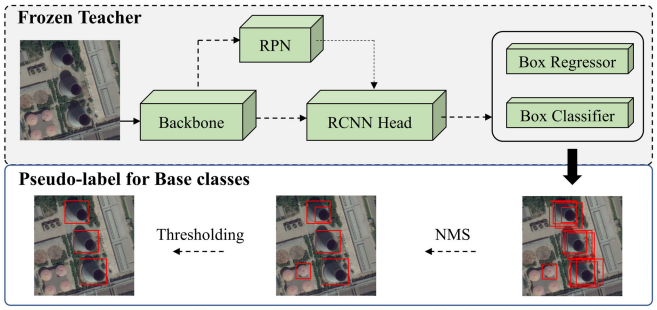


Fig. 6. Process of IOD for base classes.

1) Implicit Base Class Object Discovery: The presence of implicit base class objects can impact the original feature representation capability of the model. Hence, it is necessary to discover implicit base class objects within the data during few-shot training. On the one hand, incorporating the training of implicit base class objects can better preserve the feature representation capacity of the base classes. On the other hand, as the base classes and novel classes collectively encompass all the categories in few-shot detection, a more comprehensive representation of the base classes can aid in reducing false detections in the novel classes. To obtain the pseudolabels for these implicit base class objects, we use a teacher network, which serves the purpose of discovering these objects during the first stage of pretraining. The teacher network can provide more dependable pseudolabels for the implicit base class objects, serving as the supervision objects during few-shot training. This approach takes into consideration both the reliability of the implicit objects and the need to obtain pseudolabels for the implicit base class objects.

The process of IOD for base class objects can be divided into two main stages: first, the teacher network discovers base class objects within the training data, and then, there is postprocessing of the discovered base class objects, as depicted in Fig. 6. Considering that the performance of the teacher network deteriorates during few-shot training, the parameters of the teacher network are frozen to obtain the best results for the base classes. Since the discovered base class object boxes are usually very dense, the next step is to process these boxes. Specifically, the nonmaximum suppression (NMS) algorithm is employed to remove redundant bounding boxes. The boxes obtained by the teacher network are represented as $B = b_1, b_2, \ldots, b_n$, and the corresponding classification scores are denoted as $T = t_1, t_2, \ldots, t_n$. For the object box M with the highest classification score, the Intersection over Union (IoU) is calculated between it and other object boxes of the same class, represented as b_i. Boxes with an IoU lower than threshold N_t are retained. A low IoU between two bounding boxes indicates minimal overlap, suggesting that they likely

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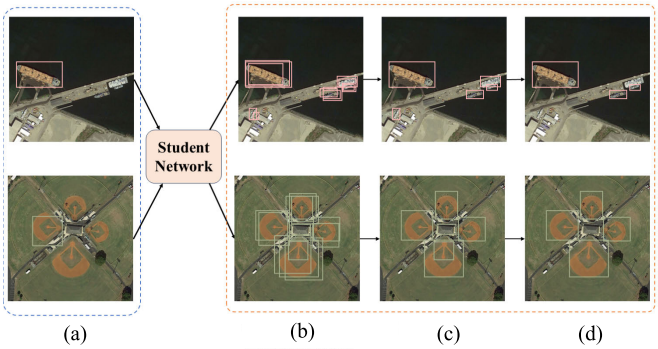


Fig. 7. Process of IOD for novel classes. (a) Few-shot annotations. (b) Predict Results of the student network. (c) Predict results after NMS. (d) Predict results after thresholding.

represent different objects and should be kept. Conversely, a high IoU between two bounding boxes implies that they belong to the same object, and the box with the lower score is suppressed. The NMS can be mathematically expressed as follows:

$$t_i = \begin{cases} t_i, & \text{IoU}(M, b_i) < N_t \\ 0, & \text{IoU}(M, b_i) \geq N_t. \end{cases} \tag{6}$$

Next, to ensure the reliability of the obtained pseudolabels, further refinement is applied to the classification scores of the results after NMS using a predefined threshold. Lower classification scores are more susceptible to classification errors, so it is important to retain objects with higher classification scores to obtain higher reliability and higher quality pseudolabels. For a score threshold σ_t, objects with scores higher than σ_t are considered more reliable, while those with scores lower than σ_t are discarded. In the experiments, the threshold σ_t is set to 0.8. The base class objects after NMS are represented as $y = y_1, y_2, \ldots, y_n$, and the pseudolabels of the processed implicit base class objects are denoted as y_t. The thresholding process can be mathematically expressed as

$$y_t = \begin{cases} y_i, & t_i \geq \sigma_t \\ 0, & t_i < \sigma_t. \end{cases} \tag{7}$$

2) Implicit Novel Class Object Discovery: The process of implicit novel class object discovery is slightly different from the process of implicit base class object discovery. In the fine-tuning stage, the student network is fine-tuned on images that provide only K instances of the novel classes and base classes, and pseudolabels for the novel class objects are generated in this stage, as shown in Fig. 7.

First, the student network using the teacher network weight initialization parameters does not have the ability to discover novel class instances when it is not trained with the few-shot dataset. Inspired by pseudolabeling techniques in semisupervised learning, we perform warm-up training on the student network for 500 iterations using the few-shot dataset, enabling it to identify implicit novel class objects. Second, the student network is used to predict the implicit novel class objects for the images [e.g., Fig. 7(a)] in the few-shot dataset, and the predicted redundant implicit objects [see Fig. 7(b)] are removed using NMS to obtain candidate object boxes. The NMS process for the results discovered by the student network can be represented as follows:

$$s_j = \begin{cases} s_j, & \text{IoU}(N, b_j) < N_s \\ 0, & \text{IoU}(N, b_j) \geq N_s \end{cases} \tag{8}$$

where b_j represents the bounding boxes discovered by the student network, s_j represents the classification scores, N_s is the NMS threshold for the student network, IoU denotes the calculation of IoU, and N represents the box with the highest classification score. After the NMS process, the results [e.g., Fig. 7(c)] obtained by the student network contain both base classes and novel class objects. The high-quality pseudolabels for the base class objects have already been generated by the teacher network, so the student network only needs to obtain pseudolabels for the novel classes. Finally, the novel classes' object boxes are filtered according to the threshold σ_s of classification scores to obtain high-confidence objects as pseudolabels [e.g., Fig. 7(d)]. The process of generating pseudolabels for the novel classes can be represented by the following equation:

$$y_s = \begin{cases} y_j, & s_j \geq \sigma_s, \ c = \text{novel} \\ 0, & \text{otherwise} \end{cases} \tag{9}$$

where s_j represents the classification scores of the bounding boxes, σ_s is the threshold for filtering the classification scores, c represents the class of the bounding box, y_j corresponds to the bounding boxes with classification scores, and y_s represents the pseudolabels for the novel classes.

F. Implicit Object Fusion

Due to the utilization of different models for generating labels for implicit objects of base classes and novel classes, merging them with existing labels is essential for training the student network. In the few-shot training phase, the pseudolabel consists of the discovered base classes and the novel class objects. Due to the overlap between pseudolabels and real labels and the fact that pseudolabels are not as reliable as real labels, it is not feasible to directly merge pseudolabels and real labels to train the student network. To address this challenge, preserving the priority of real labels while filtering out pseudolabels with significant overlaps is paramount. The objective of fusion implicit objects is to unify pseudolabels for base classes y_t, pseudolabels for novel classes y_s, and the real labels y', thereby creating labels suitable for supervised training in FSOD.

The fusion process can be summarized as follows. Initially, the classification scores of the bounding boxes within the real labels y' are assigned a value of 1. Subsequently, the pseudolabels for novel classes y_s and base classes y_t are merged with the real labels y' to obtain the fused labels y. Finally, the fused label y is processed by NMS to eliminate redundant bounding boxes and obtain the supervised label y_f. By setting the classification scores of the real labels to 1, it is ensured that the real labels have the highest confidence scores and are not removed by NMS.

The fused result y_f derived from implicit objects contains both novel classes and base classes supervision information,

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TABLE I
DIFFERENT SPLITS ON THE DIOR DATASET

SPLIT	Novel Classes	Base Classes
1	baseball field, basketball court, bridge, chimney, ship	rest classes
2	airplane, airport, expressway-toll-station, harbor, ground track field	rest classes
3	dam, golf field, storage tank, tennis court, vehicle	rest classes
4	expressway-service-area, overpass, stadium, train station, windmill	rest classes

complementing the information provided by the real labels y'. During student network training, the RPN and RCNN components are supervised using y_f. Since the model uses implicit objects as supervisory information, the negative effect of implicit objects is effectively reduced. Finally, the overall loss function for this approach is defined as follows:

$$\mathbb{L} = \mathcal{L}_{\text{RPN}}(y_p, y_f) + L_{\text{RCNN}}(y'_p, y_f) + \lambda L(y^o, y^p) + \alpha\varphi(y^T, y^S) \tag{10}$$

where y_p denotes the outputs of the RPN and y'_p denotes the collection of predicted categories and bounding boxes from the RCNN head. $L_(y^o, y^p)$ is the CMSP loss [11] in the student network. $\varphi(y^T, y^S)$ represents the MFKD loss. The parameters λ and α are set consistently with MSOCL [11].

IV. EXPERIMENT

In this section, we will introduce the experimental datasets and evaluation metrics. Subsequently, we will provide details on the specific settings of the proposed method and data configuration in the experiments. Finally, quantitative experiments, qualitative experiments, and ablation experiments will be conducted to validate the effectiveness of the proposed method.

A. Datasets and Evaluation Metrics

The DIOR dataset is a large-scale optical remote sensing image object detection dataset [56]. It consists of 5862 training images, 5863 validation images, and 11 738 testing images, totaling 23 463 optical remote sensing images. The images have a pixel size of 800×800. The dataset covers 20 categories of objects.

The NWPU VHR-10.V2 [57] is a very high-resolution remote sensing object detection dataset. It has 800 images collected from Google Earth and ISPRS Vaihingen datasets and contains ten classes of objects.

In the experiments, the mean average precision (mAP) is used to evaluate the accuracy. In order to provide a fair comparison with the frontier remote sensing FSOD methods, the calculation of mAP follows the Pascal VOC2007 benchmark. The evaluation metric can be represented as follows:

$$\text{mAP} = \frac{1}{N}\sum_{c}\int_0^1 P(R)d(R) \tag{11}$$

where $P(R)$ represents the precision–recall (PR) curve, P represents precision, R represents the recall, c represents the class, and N represents the number of classes evaluated.

B. Experimental Settings

To evaluate the effectiveness of the proposed method in few-shot scenarios, we split the 20 categories in the DIOR dataset into 15 base classes and five novel classes. Following previous work [11], [36], we divide the 20 DIOR dataset categories into four distinct sets with nonoverlapping novel classes (NOVEL 1–4) and base classes (BASE 1–4), as shown in Table I. For the NWPU VHR dataset, we split it into seven base classes and three novel classes. For a fair comparison, we follow the class split in [58]. The novel classes are *airplane, baseball diamond, and tennis court*, and the rest are base classes. We performed three replications of the proposed method and the comparison method and reported the average results.

The proposed method is implemented based on Detectron2 and PyTorch. The feature extractors for both the student and teacher networks are ResNet-101 with an FPN structure. The model training optimizer uses SGD with a momentum of 0.9 and a weight decay coefficient of 1e−4. The initial learning rate is set to 0.01 for base classes training and 0.001 for novel classes training, and they both decrease by a factor of 10 at 80% iterations. For the DIOR dataset, the base classes training totals 4000 iterations, and the novel classes training totals 9000 iterations. For the NWPU VHR dataset, the total numbers of iterations are set to 9000 and 3000.

C. Experimental Results and Analysis

To evaluate the effectiveness of the proposed method and to measure the FSOD capabilities of different methods, we conducted an experimental comparison of three classes of methods: fine-tuning-based methods (TFA [13] and DeFRCN [14]), model improvement-based methods (FSCE [20], MDFC [46], and SAE-FSDet [60]), and metric learning-based methods (RepMet [16] and AMTN [59]).

1) Quantitative Results for Novel Classes on the DIOR Dataset: The quantitative results of novel classes on the DIOR dataset are shown in Table II. The proposed method outperforms the methods based on metric learning (RepMet) and fine-tuning (TFA) in all the NOVEL 1–4 settings. The proposed method also outperforms MDFC across different settings in NOVEL 1–4, with an average improvement of 6.7%, 6.5%, and 1.2%. Notably, the proposed method exhibits the highest accuracy for novel classes in NOVEL 1, achieving an average improvement of 6.8% over the FSCE method. Furthermore, the proposed method maintains a performance advantage for novel classes over FSCE in NOVEL 2–4

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TABLE II
FSOD QUANTITATIVE RESULTS FOR THE NOVEL CLASSES ON THE DIOR DATASET. THE BEST TWO PERFORMANCES ARE HIGHLIGHTED WITH RED AND BLUE, RESPECTIVELY

Method	Year	Novel 1 (%)				Novel 2 (%)				Novel 3 (%)				Novel 4 (%)			
		3-shot	5-shot	10-shot	20-shot	3-shot	5-shot	10-shot	20-shot	3-shot	5-shot	10-shot	20-shot	3-shot	5-shot	10-shot	20-shot
Faster RCNN [23]	2015	14.0	15.4	22.8	31.1	8.2	13.4	17.9	25.2	9.1	10.5	15.4	24.5	10.6	15.0	21.8	29.0
RepMet† [16]	2019	11.3	11.3	12.1	12.2	5.2	5.6	5.9	6.8	5.5	5.7	7.4	7.8	10.9	11.6	12.8	16.7
FsDet [19]	2019	21.9	26.1	30.1	31.6	12.9	14.5	19.4	21.7	14.4	18.3	22	25.0	11.9	13.6	17.5	21.7
MPSR [34]	2020	14.8	20.9	24.5	33.1	12.9	16.2	22.7	28.8	14.7	15.9	21	24.7	15	16.7	22.2	29.3
FSCE [20]	2021	24.2	30.8	37.2	42.1	15.1	18.2	23.3	27.4	12.8	17.0	21.5	27.3	12.5	17.5	23.8	28.8
P-CNN† [36]	2021	18.0	22.8	27.6	29.6	14.5	14.9	18.9	22.8	16.5	18.8	23.3	28.8	15.2	17.5	18.9	25.7
FADI [15]	2021	16.0	17.9	31.1	37.7	12.7	13.3	22.8	27.9	14.4	18.5	23.7	29.5	14.7	15.0	23.1	28.3
DeFRCN [14]	2021	17.7	19.8	24.5	26.1	6.2	7.1	8.4	10.7	4.7	5.3	8.1	11.2	8.6	9.2	12.2	12.2
TFA [13]	2020	10.7	19.9	26.7	28.6	5.4	7.6	8.6	10.7	4.3	7.6	15.3	19.6	4.7	5.9	11.7	11.8
MDFC [46]	2022	25.5	30.1	33.0	35.7	12.6	15.7	18.7	20.6	14.6	17.8	21.6	25.4	16.2	20.2	25.1	28.0
MSOCL [11]	2022	28.5	34.6	40.8	43.3	16.8	20.6	23.6	29.2	19.6	21.4	26.0	30.2	15.6	19.0	24.6	29.4
AMTN† [59]	2024	22.0	24.7	29.5	32.2	15.1	18.1	20.6	24.7	18.5	20.5	23.7	29.2	-	-	-	-
SAE-FSDet† [60]	2024	28.8	32.4	37.2	42.7	14.0	15.7	17.4	21.3	16.7	19.1	**28.4**	29.9	**17.3**	20.5	22.7	26.8
MFKDIOD(Ours)		**33.2**	**37.7**	**41.7**	**43.5**	**20.0**	**21.8**	**28.8**	**29.5**	**21.6**	**24.9**	27.6	**31.5**	16.7	**21.0**	**25.8**	**30.6**

TABLE III
FSOD QUANTITATIVE RESULTS FOR THE BASE CLASSES ON THE DIOR DATASET. THE BEST TWO PERFORMANCES ARE HIGHLIGHTED WITH RED AND BLUE, RESPECTIVELY

Method	Year	Novel 1 (%)				Novel 2 (%)				Novel 3 (%)				Novel 4 (%)			
		3-shot	5-shot	10-shot	20-shot	3-shot	5-shot	10-shot	20-shot	3-shot	5-shot	10-shot	20-shot	3-shot	5-shot	10-shot	20-shot
Faster RCNN [23]	2015	42.9	47.1	51.8	56.4	41.4	43.2	46.6	51.1	45.2	49.6	51.6	56.7	38.2	40.2	45.5	50.4
RepMet† [16]	2019	47.8	49.4	49.3	48.6	49.6	51.1	51.2	50.4	48.7	49.6	52.3	52.4	47.4	49.2	50.5	53.5
FsDet [19]	2022	54.6	57.5	58.7	59.8	55.6	55.9	58.8	59.9	57.1	58.2	60.4	62.4	52.0	53.5	54.2	55.2
MPSR [34]	2020	44.1	47.6	47.8	51.6	43.2	44.0	44.2	46.5	47.3	47.3	49.3	51.4	41.7	39.7	42.5	44.3
FSCE [20]	2021	44.6	54.8	60.0	61.3	39.8	47.2	52.8	55.7	49.9	55.1	60.7	60.0	51.3	44.6	50.7	55.8
P-CNN† [36]	2021	47.0	48.4	50.9	52.2	48.9	49.1	52.5	51.6	49.5	49.9	52.1	53.1	49.8	49.9	51.7	52.3
FADI [15]	2021	54.6	54.4	52.7	55.2	52.4	51.6	47.9	51.7	55.6	55.7	50.2	52.5	52.7	52.8	46.7	48.5
DeFRCN [14]	2021	**65.9**	66.0	66.0	66.1	64.9	65.0	65.1	65.3	66.4	66.6	66.8	67.1	62.6	62.7	63.4	63.7
TFA [13]	2020	52.8	56.5	57.8	59.6	50.2	52.4	53.5	55.3	53.7	58.3	60.4	61.4	49.3	52.6	54.0	55.8
MDFC [46]	2022	63.3	64.5	64.3	64.5	62.6	63.9	63.2	63.5	65.3	66.1	66.2	66.4	64.0	64.6	64.7	64.5
MSOCL [11]	2022	50.2	56.6	63.1	64.5	47.6	52.0	56.0	58.2	55.4	57.9	60.3	63.5	50.3	47.3	51.6	56.0
AMTN† [59]	2024	45.5	47.6	49.7	52.0	48.1	50.0	51.1	52.3	47.1	48.5	51.7	52.0	-	-	-	-
MFKDIOD(Ours)		65.7	**67.1**	**68.8**	**69.9**	**66.0**	**66.5**	**68.0**	**69.6**	**67.5**	**68.0**	**69.6**	**70.3**	**64.3**	**67.1**	**68.1**	**69.1**

settings. This is because the proposed method discovers novel class implicit objects in the image, which reduces the misguidance of incomplete novel class labels; moreover, the model learns the features of the novel classes more accurately, guided by the knowledge of the base class features. Furthermore, compared to the baseline method MSOCL, the proposed method achieves an average accuracy improvement of 2.2%, 2.0%, and 1.5% for the four sets of novel classes in different shot settings. In addition, it can be observed that as the shot increases, the growth rate of the proposed method decreases. This is mainly due to the decrease in the occurrence of implicit novel class objects as the number of instances in the few-shot

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TABLE IV
FSOD QUANTITATIVE RESULTS FOR THE BASE CLASSES AND NOVEL CLASSES ON THE NWPU VHR-10 DATASET

Method	3-shot (%)		5-shot (%)		10-shot (%)		20-shot (%)	
	nAP	bAP	nAP	bAP	nAP	bAP	nAP	bAP
TFA	17.6	**87.6**	19.5	**88.5**	22.3	88.4	26.1	89.4
MFDC	16.7	42.4	21.7	42.9	35.1	42.2	38.5	42.4
DeFRCN	31.0	84.3	36.4	86.7	38.1	87.3	49.0	88.1
MSOCL	53.0	84.0	62.1	86.9	70.7	86.7	76.7	87.4
OURS	**58.2**	86.3	**69.5**	87.9	**73.1**	**88.7**	**78.3**	**89.7**

nAP denotes the AP50 of the novel classes.
bAP denotes the AP50 of the base classes.

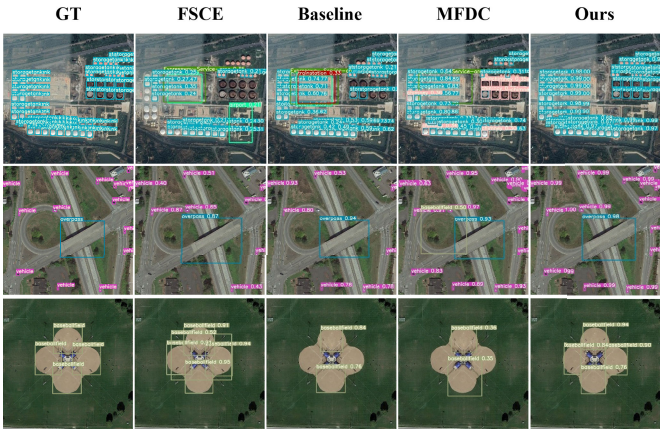


Fig. 8. Qualitative results in the proposed method and the comparison method on the DIOR dataset under SPLIT 1 and five-shot settings.

training increases. This also indicates that the inconsistency between the number of instances and the number of labels in the images is more likely to occur with fewer samples and indirectly reflects the existence of implicit novel class objects.

2) Quantitative Results for Base Classes on the DIOR Dataset: To validate the proposed method's effectiveness in preserving the feature representation capacity of base classes, Table III reports the quantitative results of base classes on the DIOR dataset for different few-shot detection methods. The proposed method demonstrates significant advantages compared to the TFA and the RepMet. In BASE 1, the proposed method achieves performance improvements of 12.9%, 10.6%, 11.0%, and 10.3% over the TFA method. Compared to the best-performing few-shot detection method DeFRCN, the proposed method achieves an average improvement of 1.6%, 1.5%, and 4.0% in BASE 1–4, respectively. In contrast to FSCE, the proposed method exhibits improvements of 16.9%, 13.9%, and 18% in BASE 1–4. In conclusion, the results show that the proposed method achieves advanced performance for base classes compared to the state-of-the-art method. On the one hand, this indicates that the discovery of implicit base class objects plays a crucial role in maintaining the performance of base classes. On the other hand, it suggests that the interference from implicit base class objects during few-shot training affects the feature representations and performance of base classes. Compared to the baseline method, the accuracy of BASE 1–4 is improved by an average of 14.9%, 8.16%, and 13.34%, respectively. These results demonstrate a significant performance boost of the proposed method over the baseline method in terms of base classes. This is due to the proposed method incorporating labels of implicit base class objects, providing more complete annotations for base class objects. In addition, the base class feature knowledge from the teacher network leads to more comprehensive base class feature representations. As a result, the proposed method exhibits pronounced advantages in maintaining base class performance.

3) Quantitative Results on the NWPU VHR-10 Dataset: We also conducted comparison experiments on the NWPU VHR-10 dataset, and the results are shown in Table IV, where our method achieves first or second place in all accuracies. For the baseline method MSOCL, although the accuracy on the novel classes exceeds the other compared methods, it does not have an advantage on the base classes. After the improvement of MFKD and IOD modules, the proposed method not only better preserves the performance of the base classes but also further improves the novel classes. With only five instance-level samples required, the proposed method can achieve 69.5% and 87.9% mAP on the novel classes and base classes, respectively.

4) Qualitative Results: Qualitatively, Fig. 8 visualizes the detection results of FSCE, the baseline method MSOCL, MFDC, and the proposed method in the SPLIT 1 and five-shot settings on the DIOR dataset. In the first row, where storage tanks from the base classes are presented, FSCE and the baseline method display evident misclassifications and omissions. MFDC demonstrates fewer omissions but still exhibits misclassifications for storage tank objects. In contrast, the proposed method delivers robust detection results for densely packed base class objects that align more closely with the ground-truth labels. In the second row, focusing on base class vehicle objects, both FSCE and the baseline method exhibit omissions, while MFDC encounters misclassifications. The proposed method not only achieves more precise category and location predictions but also provides higher confidence scores for category prediction. In the final row, the proposed method also demonstrates better results for novel class detection. Specifically, FSCE and the baseline method manifest more pronounced omissions and misclassifications in the detection results. The base classes pretrained model shows fewer missed detections. Further analysis reveals that the missed detections are mainly caused by the declining detection performance of the base class. During training, the presence of implicit objects from the base classes leads to erroneous guidance for the model, resulting in the model mistakenly classifying some implicit objects as background in the detection results. The proposed method discovers implicit objects in the training data, which helps fill in the missing labels. With relatively complete labeled training, the proposed method exhibits fewer misdetections and missed detections for both base and novel classes.

However, our method also has limitations in some specific scenarios, as shown in Fig. 9. For instance, ships with unique

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Fig. 9. Visualization of failure cases on the DIOR dataset under SPLIT 1 and five-shot settings.

TABLE V
RESULTS OF ABLATION EXPERIMENTS ON THE DIOR DATASET UNDER THE SPLIT 1 AND FIVE-SHOT SETTINGS

baseline	MFKD	IOD for base	IOD for novel	nAP (%)	bAP (%)
✓				35.35	59.10
✓	✓			38.94	61.84
✓	✓	✓		38.85	**67.11**
✓	✓		✓	**40.52**	62.27
✓	✓	✓	✓	40.34	66.99

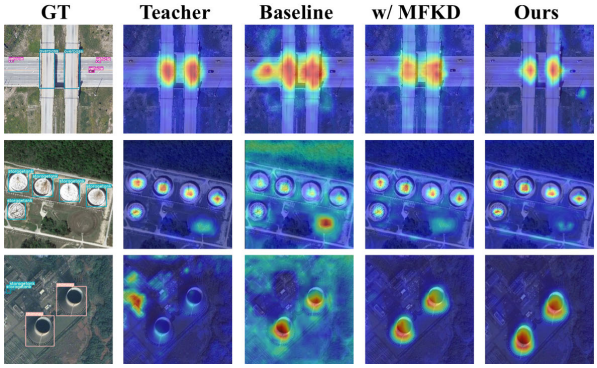


Fig. 10. Feature activation maps of different ablation models.

appearances cannot be recognized, some "bridge" objects are misclassified as "overpass," and very small ships are difficult to detect. These failure cases can be summarized into three categories: objects with significant appearance differences from the few-shot instances are hard to recognize; objects with significant appearance differences from the few-shot instances are hard to recognize; objects with high interclass similarity are prone to misclassification; and objects with indistinct features are easily missed.

D. Ablation Study

1) Ablation Study on the Proposed Modules: To analyze the specific impact of the MFKD and IOD, we performed ablation experiments under the SPLIT 1 and five-shot settings (see Table V). In addition, we visualized the features activation maps (see Fig. 10) of the different ablation models using Grad-CAM [61].

To demonstrate the phenomenon and the effect of the MFKD module, we compare the accuracy of the baseline model and the model after using the MFKD module on the base classes (rows 1 and 2 of Table V) and the feature activation maps (the first two rows of Fig. 10). In Table V, the MFKD module improves the performance of the base classes by 2.74% compared to the 59.1% performance of the baseline method. As shown in the first two rows of Fig. 10, the baseline network (column 3) without the MFKD module shows base class feature forgetting with respect to the teacher network (column 2). After using the MFKD module, the network obtained better feature activation maps and was more similar to the teacher network. This suggests that the use of MFKD enables the student network to effectively learn the feature knowledge of the teacher network and alleviates the problem of base class feature forgetting. In addition, we found the MFKD module resulted in a 3.59% improvement in the novel classes. This finding suggests that the base class feature representations boost the model's learning of the novel classes.

To analyze the effects of the IOD module, we compared the accuracy before and after using the IOD module. As shown in rows 3–5 of Table V, when implicit base class object discovery (IOD for base) is used alone, the accuracy of the base classes can be improved by 5.27%. When implicit novel class object discovery (IOD for novel) is used alone, the accuracy of the base classes can be improved by 1.58%. The best overall performance was achieved when both novel classes and implicit base class object discovery were used, with a 4.99% improvement in novel classes and a 7.89% improvement in base classes compared to the baseline method.

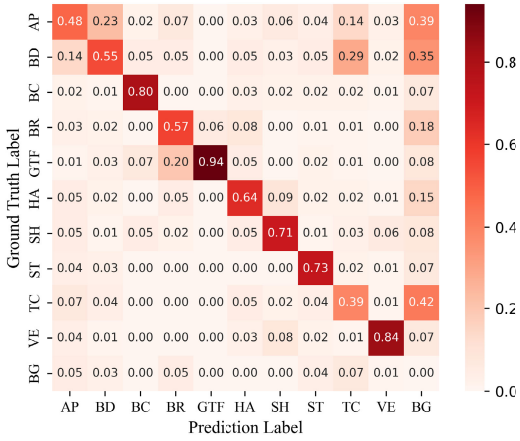
To analyze the effects of the IOD module's generation of pseudolabels, we visualized the feature activation maps before and after using the IOD module, as shown in Fig. 10 (last two columns). After using the IOD module for the novel classes, the response of the base and novel classes activation feature map is more focused on the objects.

Besides the quantitative analysis, we also visualized the confusion matrices on the NWPU-VHR dataset with and without the IOD module, as shown in Fig. 11. From the two confusion matrices, it can be seen that without the IOD module, more instances of the novel classes (AP, BD, and TC) are misclassified into the background due to the misdirection of implicit objects. With the help of the IOD module, the misclassification of novel and base classes is significantly reduced.

2) Analysis of Multiscale Features: The MFKD extracts knowledge from multiscale features as soft labels to guide the student network. Therefore, we performed ablation experiments on distilling features at different scales (see Table VI). It can be observed that when using features of different scales, the best overall performance is achieved with P2, and the performance gradually decreases from P2 to P5, with P5 being the worst. As the number of scales used for feature distillation increases, the effectiveness of knowledge improves. Specifically, when distilling features from P2, P3, P4, and P5 layers, there is a 1.67% improvement in novel class accuracy and a 1.13% improvement in base class accuracy compared to using a single scale. The experiment results indicate that multiscale feature knowledge provides more guidance for the student network.

3) Analysis of Hyperparameters: Fig. 12 shows the accuracy of the novel classes for SPLIT 1 and five-shot settings with the distillation temperature T. The distillation

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(a)



(b)

Fig. 11. Classification confusion matrix for the MFKDIOD method with and without IOD module on the NWPU VHR-10 dataset. (a) Without the IOD module. (b) With the IOD module. The abbreviations for the categories are given as follows: AP: airplane; BD: baseball diamond; BC: basketball court; BR: bridge; GTF: ground track field; HA: harbor; SH: ship; ST: storage tank; TC: tennis court; VE: vehicle; and BG: background.

TABLE VI

RESULTS OF ABLATION EXPERIMENTS ON MULTISCALE FEATURES FOR DISTILLATION ON THE DIOR DATASET IN THE SPLIT 1 AND FIVE-SHOT SETTINGS

P2	P3	P4	P5	nAP (%)	bAP (%)
✓				37.27	60.71
	✓			36.98	<u>61.83</u>
		✓		36.84	60.01
			✓	36.13	59.34
✓	✓			37.53	60.96
✓	✓	✓		<u>37.80</u>	61.10
✓	✓	✓	✓	**38.94**	**61.84**

temperature T ranges from 0.1 to 1.0 with a sampling interval of 0.1. The result indicates that there is no linear relationship between the accuracy of the novel classes and T, showing some fluctuations. When T is greater than 0.2, the accuracy of the novel classes shows a decreasing trend; and when T is equal to 0.2, the accuracy of the novel classes reaches the highest, which is 38.94%. Therefore, the proposed method set T to 0.2 in the experiment.

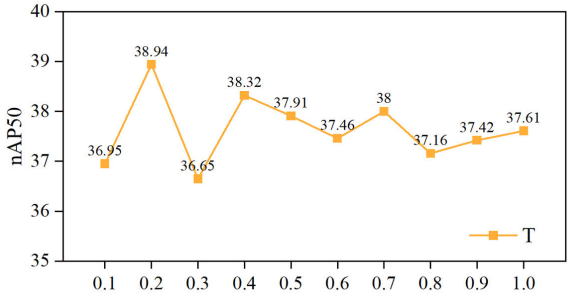


Fig. 12. Effect of distillation temperature T on the DIOR dataset.

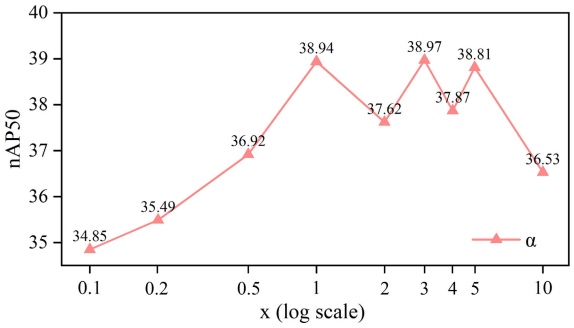


Fig. 13. Effect of distillation loss weight α on the DIOR dataset.

In order to select the hyperparameter α that is suitable for the proposed method, we have performed ablation experiments for α. The accuracy of the novel class is shown in Fig. 13. We first set α to 1 to observe the values of multiscale distillation losses and other losses, which do not differ by more than a factor of 10 between each loss except the regression loss. Therefore, we set the search range of α to 0.1 to 10 and performed sampling uniformly in log space to obtain candidate values [0.1, 0.2, 0.5, 1, 2, 5, 10]. After ablation experiments, it was found that nAP achieves the top two accuracies when α takes values of 1 and 5. Therefore, the optimal α should lie between 1 and 5. To further explore whether there is a more suitable value, we further set the parameter α to [1, 2, 3, 4, 5] for the experiment. We found that the accuracy fluctuation slows down when α lies in the interval 1–5, where the highest accuracy is obtained when α is taken to 3. Therefore, α is set to 3 in our experiments.

To investigate the impact of using different thresholds on the results, Fig. 14 displays the variation of accuracy for the base and novel classes at different σ_t's and σ_s's, respectively. The teacher network is insensitive to the threshold σ_t due to the fact that the teacher network has already learned sufficiently in the pretraining phase. In order to maintain the performance of the base class and minimize the false detection of implicit objects in the base class, σ_t is set to 0.8 (a second-best value for accuracy). The student network shows an increasing and then decreasing trend as the threshold σ_t increases. This is because when σ_s is low, it brings very many false detections; when σ_s is high, it is easy to miss implicit objects. Therefore, the threshold σ_s for the student network was set to 0.85.

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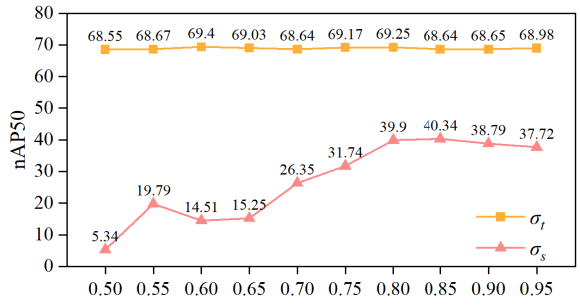


Fig. 14. Effect of different thresholds on the DIOR dataset.

TABLE VII
MODEL PARAMETER SIZE, FLOATING-POINT OPERATIONS, AND INFERENCE SPEED

Method	Backbone	Parameters(M)	FLOPs(G)	FPS(frame/s)
TFA	ResNet-101	60.3	166.082	25.9
MSOCL	ResNet-101	60.9	166.082	24.5
DeFRCN	ResNet-101	52.1	352.549	16.5
MFDC	ResNet-101	60.5	351.675	13.4
OURS	ResNet-101	61.6	166.082	24.3

4) Analysis of Model Efficiency: Table VII presents the parameter size (parameters), the testing inference floating-point operations (FLOPs), and inference speed for different methods. For the proposed method, the parameter count is 61.6M, the FLOPs are 166.082G, and the inference speed is 24.3 frame/s. Compared to the baseline method, the FLOPs remain unchanged, while the parameter count increases by 0.7M. Compared with the fastest inference methods (TFA) and the baseline method (MSOCL), the proposed method is only slower by 1.6 and 0.2 frames/s. Overall, the proposed method is still highly efficient while guaranteeing the best accuracy.

V. CONCLUSION

In this article, we present a new remote sensing FSOD method that uses MFKD and IOD modules to solve the problem of base class feature representations forgetting and implicit object interference. Specifically, the proposed method utilizes knowledge distillation to transfer base class feature knowledge from the teacher network to the student network, continuously reinforcing the student network's base class knowledge to mitigate knowledge forgetting. In addition, we propose an IOD module to obtain more complete labeling information, which reduces misguidance from incomplete labels. To validate the effectiveness of the proposed method, we design and conduct a large number of experiments on the DIOR dataset. Moreover, through the experiments, we observe a positive correlation between the performance of base classes and the performance of novel classes. By mitigating the decay of base class knowledge, the model acquires more foundational knowledge, which facilitates the training of novel classes. We also find that implicit objects do introduce erroneous supervision during few-shot training, but as the number of training instances increases, the influence of implicit objects gradually diminishes.

In the future, we will consider the semantic relationship between categories and the feature representation capability of the remote sensing large model. Based on our proposed method, we will explore the potential of remote sensing large models in remote sensing FSOD tasks to further improve the accuracy of remote sensing FSOD.

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